THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 6: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest digit $*$ making $5*48$ divisible by 3 is: 1

(a) 2 **(b)** 0 **(c)** 1 **(d)** 3

2. The Collatz sequence from 8 (including 8 and 1) has how many terms? 1

(a) 5 **(b)** 6 **(c)** 4 **(d)** 3

3. Applying Kaprekar's routine once to 4172 gives: 1

(a) 6173 **(b)** 6176 **(c)** 6175 **(d)** 6174

4. The number of 2-digit palindromes is: 1

(a) 8 **(b)** 10 **(c)** 11 **(d)** 9

5. The sum of 5 consecutive numbers starting at 10 is: 1

(a) 59 **(b)** 61 **(c)** 62 **(d)** 60

6. For every whole number n, $n^2 + n$ is always: 1

(a) even **(b)** a multiple of 3 **(c)** odd **(d)** a prime number

7. A number has digit sum 12. It is divisible by: 1

(a) 3 but not 9 **(b)** both 3 and 9 **(c)** neither 3 nor 9 **(d)** 9 but not 3

8. Adding 72 and its reverse gives: 1

(a) 101 **(b)** 99 **(c)** 100 **(d)** 98

9. The smallest digit $*$ making $7*21$ divisible by 3 is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 1

10. The Collatz sequence from 11 (including 11 and 1) has how many terms? 1

(a) 14 **(b)** 15 **(c)** 17 **(d)** 16

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The number $5*14$ is divisible by 9. Find the digit $*$ and write the number. 2

12. The sum of three consecutive odd numbers is 51. Find them. 2

13. Use reverse-and-add on 39 and state how many steps are needed to get a palindrome. 2

14. The number $62*7$ is divisible by 9. Find the digit $*$ and write the number. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Apply Kaprekar's routine to 7621 repeatedly until you reach 6174. 3

16. The sum of a two-digit number and its reverse is 99, and its digits differ by 3. Find the number (tens digit larger). 3

17. Write 75 as the sum of 5 consecutive numbers. 3

OR

Apply Kaprekar's routine to 9810 repeatedly until you reach 6174. 3

18. The sum of a two-digit number and its reverse is 121, and its digits differ by 1. Find the number (tens digit larger). 3

OR

Write 63 as the sum of 7 consecutive numbers. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write the Collatz sequence from 7. Find the largest number reached, the number of even terms and the number of odd terms. 5

OR

List all three-digit palindromes with first digit 1 that are divisible by 3. 5

20. Write the Collatz sequence from 20. Find the largest number reached, the number of even terms and the number of odd terms. 5

OR

List all three-digit palindromes with first digit 2 that are divisible by 3. 5

END OF QUESTION PAPER — Total: 40 Marks

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